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Structure to Affinity

Predicting Human GSK-3β Binding from Molecular Structure

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Final Technical Report

1 Introduction

Glycogen synthase kinase-3 beta (GSK-3β) is a serine/threonine kinase with a structurally characterized ATP-binding pocket. It has been studied in connection with tau phosphorylation, beta-amyloid-mediated neurodegeneration, and Alzheimer disease [4, 5]. These links make GSK-3β scientifically interesting, but they do not make every measured inhibitor interchangeable. Binding affinity depends on molecular structure, assay protocol, protein construct, and the chemical domain represented in the data.

This project asks: *How accurately can experimentally measured binding affinity to human GSK-3β be predicted from a small molecule's physicochemical descriptors, functional groups, and structural fingerprint?* The response is pK_d, a logarithmic transformation of the equilibrium dissociation constant. A secondary question is whether a compact, interpretable descriptor panel performs nearly as well as a detailed fingerprint. The intended result is a single-target ligand-based quantitative structure-activity relationship (QSAR) model for compounds similar to those observed here. It is not a causal model of binding and is not intended for a different protein.

The analysis follows a pre-specified fallback in the execution plan. The preferred same-platform cohort began with 117 displayed measurements, but censoring and deduplication left only 18 unique molecules. I therefore promoted the larger exact-K_d cohort, which retained 115 unique molecules. Grouped cross-validation selected a random forest using a reduced Morgan fingerprint. On a 22-molecule scaffold-disjoint holdout, its mean absolute error (MAE) was 1.108 pK_d units, compared with 1.286 for a median baseline. This is a nominal improvement, but its paired scaffold-bootstrap interval includes zero. The result is useful primarily because it quantifies both the available predictive signal and the uncertainty hidden by an ordinary random split.

2 Dataset Description and Preparation

2.1 Source, provenance, and observational units

All measurements come from BindingDB, a public database of experimentally measured protein-small-molecule affinities [1]. The target is human GSK-3β, BindingDB polymer 1003 and UniProt P49841. The execution plan specified the target TSV at

`https://www.bindingdb.org/rwd/data/downloads/ptenK0/BDBpoly_1003.tsv`.

The endpoint-specific query is

`https://www.bindingdb.org/rwd/jsp/dbsearch/PrimarySearch_ki.jsp?polymerid=1003&column=Kd&submit=Search&tag=polkd`.

BindingDB defines one TSV row as one reported ligand-target binding measurement; each row can include a ligand SMILES string, affinity endpoint, target identity, provenance, and external identifiers [2]. The raw observational unit is therefore a measurement, while the final modeling unit is a standardized unique molecule.

The live database views were not internally synchronized. The frozen target TSV contains 3,870 rows and 640 columns but only 58 nonblank K_d fields. The public endpoint query displays 252 K_d records. I preserved the 10.5 MB TSV and all six result pages, recorded SHA-256 hashes, and constructed a 252-row audit table from the public results. Every constructed row retains a unique BindingDB reaction ID,

SMILES, displayed K_d, assay description, DOI or PMID when available, institution, and reaction URL. The snapshot date is August 13, 2026. BindingDB-curated data are CC BY 3.0; ChEMBL-imported records retain CC BY-SA 3.0 terms [3].

2.2 Cohorts and attrition

The candidate primary cohort consists of the Karaman et al. and Davis et al. kinase panels, which contribute 45 and 72 displayed measurements, respectively [6, 7]. Both use the Ambit active-site competition-binding platform. The extended cohort contains all 252 displayed human P49841 K_d records.

Table 1 shows why the primary cohort was not modeled. BindingDB reports 91 of its 117 values as bounds, mainly `>1.00E+4 nM`. Treating these bounds as exact labels would fabricate response values. Only 26 exact positive measurements remain, and repeated compounds reduce this to 18 standardized molecules. The execution plan required at least 80 molecules and 10 scaffold groups for the primary cohort. It therefore fails even though it has 17 scaffold groups. The extended cohort retains 115 molecules across 74 scaffolds and clears the fallback minimum of 50.

Table 1. Raw-to-model attrition. Exact-value filtering, rather than structure parsing, causes most sample loss.

Stage	Primary same-platform	Extended exact-K_d
Displayed K_d measurements	117	252
Exact positive K_d measurements	26	141
RDKit-parseable measurements	26	141
Unique standardized molecules	18	115
Nonempty scaffold groups	17	74

2.3 Cleaning and derived response

I retained exact positive K_d measurements in nM and excluded interval-censored values. RDKit parsed each SMILES, applied molecular cleanup and `FragmentParent` to remove disconnected counterions, and generated canonical isomeric SMILES and InChIKeys [8]. Stereochemistry and formal charge were retained when possible. All 141 exact rows parsed successfully.

Repeated standardized molecules were combined before splitting. The target for molecule i is the median

$$pK_{d,i} = -\log_{10}(K_{d,i}[\mathrm{M}]) = 9 - \log_{10}(K_{d,i}[\mathrm{nM}])\,. \tag{1}$$

Higher pK_d denotes stronger binding. Fourteen molecules had multiple exact measurements; two had replicate ranges above 1.0 pK_d unit and were flagged for sensitivity analysis. Provenance fields, database IDs, assay text, and scaffold labels were retained for auditing but never used as predictors.

3 Exploratory Data Analysis

The final target ranges from 3.21 to 10.34 pK_d with a median of 5.89. This corresponds to a chemically wide affinity range, spanning more than seven orders of magnitude. Figure 1 also shows that the scaffold distribution is highly concentrated: one scaffold contains 29 of 115 molecules, while most appear once.

This concentration directly motivates scaffold-disjoint validation. A random row split could place close analogs from the dominant chemical series in both training and test data.

Figure 1. The exact-value filter, not SMILES parsing, drives sample loss

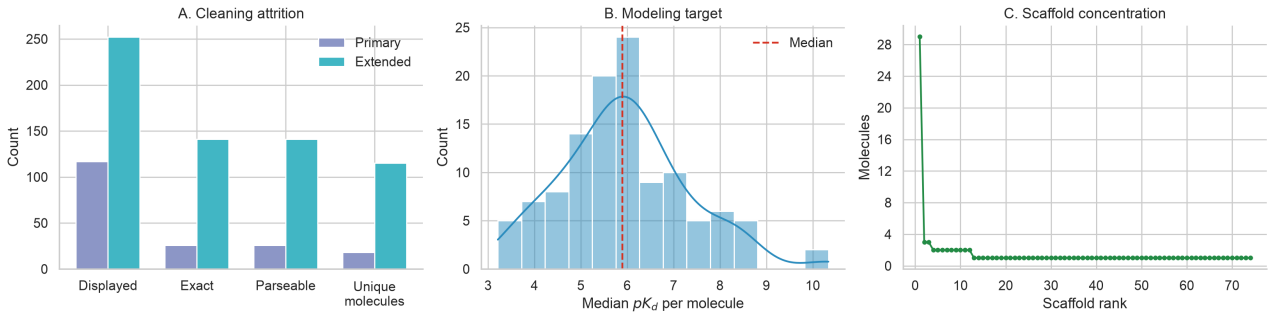


Figure 1. Data audit and target structure. Panel A shows the collapse of the primary cohort after exact-value filtering. Panel B shows the broad pK_d distribution. Panel C shows the long-tailed scaffold counts.

I next examined molecular weight, calculated logP, topological polar surface area, fraction sp^3, ESOL predicted solubility, and ring count against pK_d. The individual Spearman associations were weak, and smooth quadratic trends were not consistently monotone (Figure 2). This argues against a single-descriptor explanation and supports the planned multivariate comparison. It also limits interpretation: apparent feature effects may reflect correlations among chemical series rather than an isolated physical mechanism.

Figure 2. Individual descriptors show weak, nonlinear associations with affinity

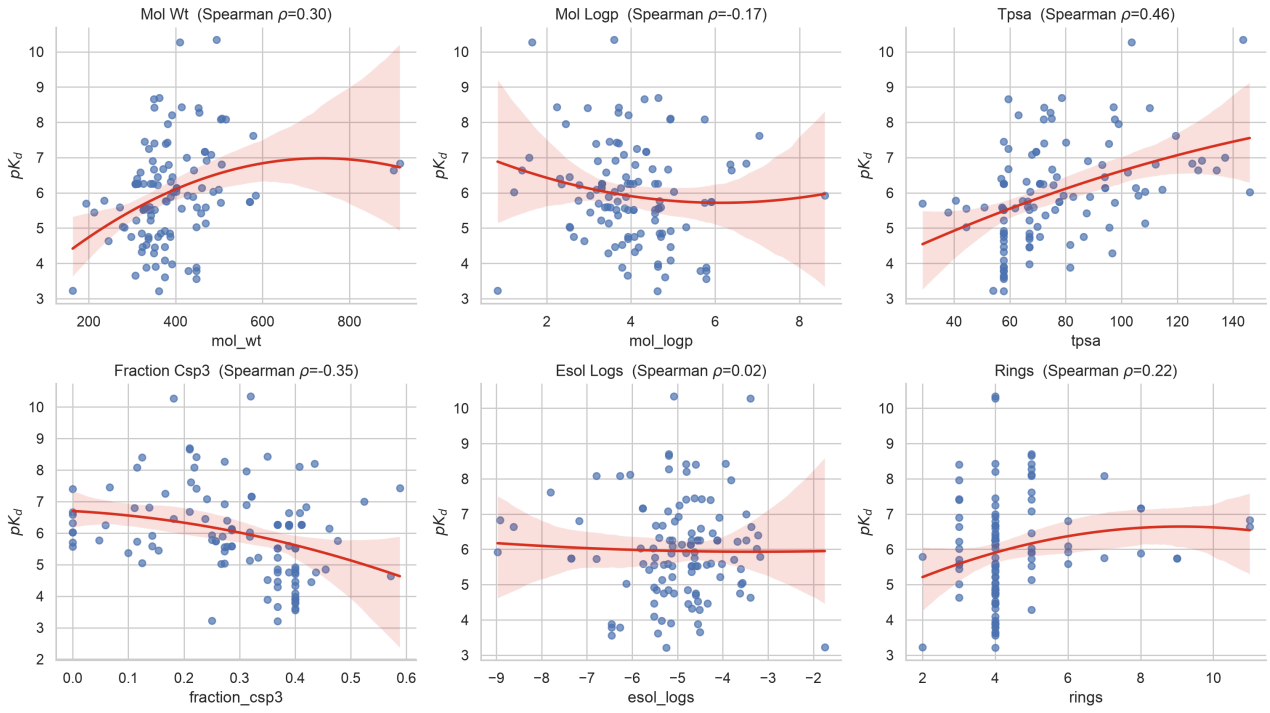


Figure 2. Selected descriptor-response relationships. No scalar descriptor provides a strong standalone ranking of affinity. Curves are descriptive quadratic fits, not causal dose-response functions.

4 Project Design and Implementation

4.1 Molecular representations

The interpretable representation contains 27 features: molecular weight, heavy-atom and heteroatom counts, formal charge, Wildman-Crippen logP, topological polar surface area, hydrogen-bond donors and acceptors, rotatable bonds, total/aromatic/aliphatic rings, fraction sp^3, Delaney ESOL predicted log-solubility, and 13 functional-group counts [9, 10, 11]. ESOL is explicitly a calculated estimate, not a measured solubility endpoint.

The structural representation is a 2,048-bit, radius-2 Morgan fingerprint with chirality [12]. The combined representation concatenates the descriptor panel and fingerprint. In response to the execution-plan feedback about high dimension, each fingerprint pipeline removes constant bits and selects the 64 bits with the strongest univariate training-fold association. This step occurs inside the cross-validation pipeline. The selector never sees its validation fold or the final holdout.

4.2 Models and hyperparameter selection

The baseline is a `DummyRegressor` that predicts the training-fold median. Ridge regression uses median imputation, standardization, and $\alpha \in \{0.1, 1, 10, 100\}$. Random forests search 300 or 600 trees, maximum depth in `{None, 5, 10}`, minimum leaf size in `{1, 3, 5}`, and `max_features` in `{sqrt, 0.5}`. Ridge and forest are each evaluated with descriptors, fingerprint, and combined inputs.

4.3 Leakage-resistant evaluation

Bemis-Murcko scaffolds define the split groups [13]. A seeded group shuffle holds out 17 scaffolds containing 22 molecules; the remaining 57 scaffolds and 93 molecules are used for model selection. The two sets share neither a molecule nor a scaffold. Five-fold `GroupKFold` controls all preprocessing, feature selection, representation choice, and hyperparameter selection. The final holdout is evaluated only after the selected configuration is fixed.

The primary metric is MAE in pK_d. Because an error of e pK_d units corresponds to approximately a 10^e-fold affinity error, MAE remains interpretable on the laboratory scale. Secondary metrics are root mean squared error (RMSE), R^2, and Spearman rank correlation. Final uncertainty is computed from 2,000 bootstrap samples of whole holdout scaffolds rather than individual molecules.

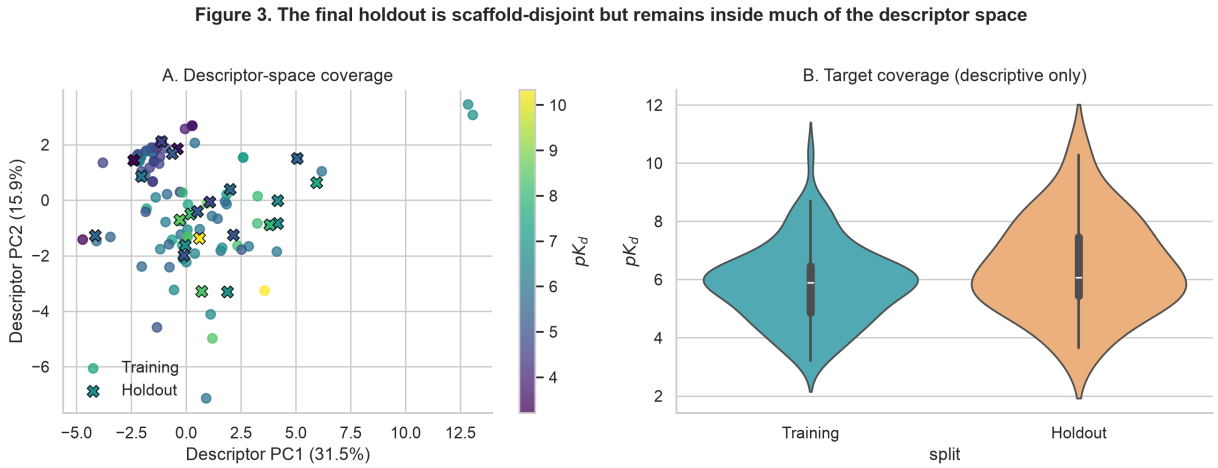


Figure 3. Frozen split coverage. PCA is fit only on standardized training descriptors, then applied to the holdout. Color shows pK_d for descriptive coverage assessment; it is not used to choose the split.

5 Results

5.1 Grouped model comparison

Table 2 and Figure 4 summarize model selection. The fingerprint random forest has the lowest grouped-CV MAE, 0.889 ± 0.234 pK_d. Its selected parameters are 600 trees, depth 10, minimum leaf size 1, and square-root feature sampling. Every learned configuration improves on the median baseline in mean CV MAE, but all mean fold-level R^2 values are negative. This is not inconsistent with an MAE improvement: R^2 is unstable in small, chemically uneven folds and compares squared error with each fold's local mean.

Table 2. Five-fold scaffold-grouped cross-validation after within-fold tuning. Lower MAE and RMSE are better.

Model	Representation	MAE ± SD	RMSE	R^2	Spearman
Random forest	Fingerprint	0.889 ± 0.234	1.131	-0.452	0.199
Random forest	Combined	0.925 ± 0.214	1.145	-0.529	0.258
Ridge	Combined	0.945 ± 0.161	1.173	-0.518	0.126
Ridge	Fingerprint	0.962 ± 0.187	1.173	-0.510	0.106
Random forest	Descriptors	0.966 ± 0.196	1.173	-0.654	0.226
Ridge	Descriptors	0.967 ± 0.221	1.193	-0.629	0.126
Median baseline	None	1.116 ± 0.375	1.383	-1.121	–

The secondary representation question has a qualified answer. The best descriptor-only configuration reaches 0.966 MAE, only 0.077 pK_d behind the fingerprint forest. That gap is smaller than the fold-to-fold standard deviations. The compact panel therefore approaches the fingerprint in this sample, but the fingerprint still wins the pre-specified selection. Adding descriptors to the fingerprint does not improve the selected forest, suggesting that the scalar summaries add little independent signal after local substructures are represented.

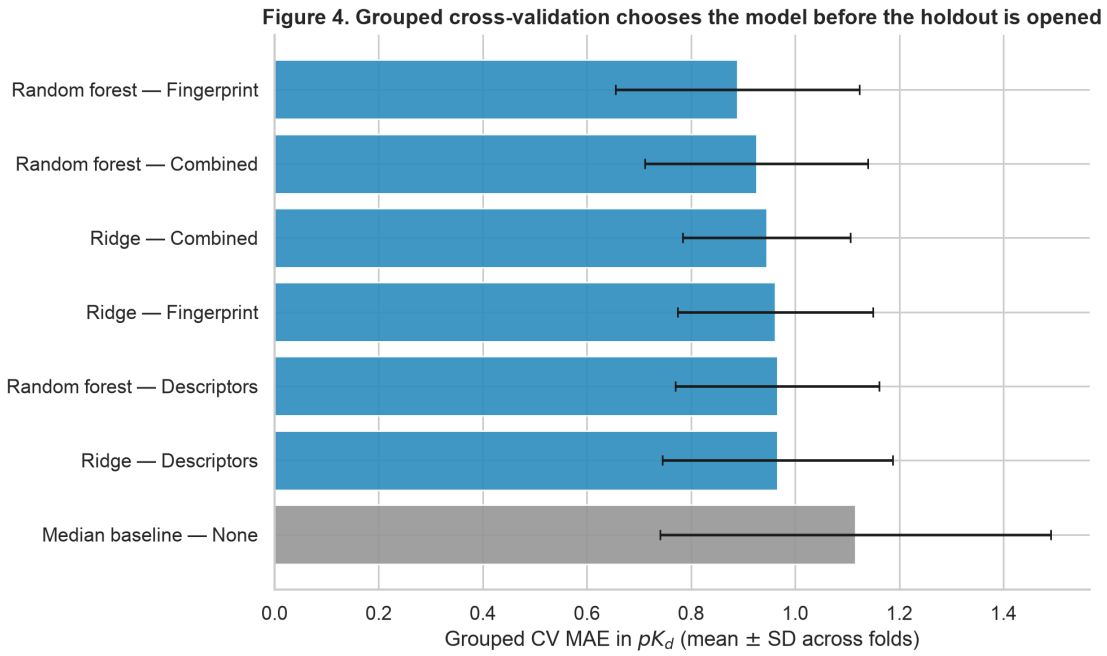


Figure 4. Grouped-CV MAE. Error bars are standard deviations across the five scaffold folds, not confidence intervals.

5.2 Final scaffold holdout

On the untouched holdout, the selected model reaches MAE 1.108 pK_d, RMSE 1.404, $R^2 = 0.210$, and Spearman $\rho = 0.448$ (Table 3). The MAE corresponds to a typical affinity error of about $10^{1.108} = 12.8$-fold. The median baseline has MAE 1.286 and negative R^2. The learned model therefore meets the execution plan's point-estimate success criterion.

Table 3. One-time scaffold-disjoint holdout results. Parentheses give 95% scaffold-bootstrap intervals.

Model	MAE	RMSE	R^2	Spearman
Median baseline	1.286 (0.807, 1.819)	1.677 (1.061, 2.227)	-0.127	–
Random forest, fingerprint	1.108 (0.772, 1.490)	1.404 (0.922, 1.884)	0.210	0.448

The paired MAE improvement is 0.178 pK_d. Its scaffold-bootstrap 95% interval is -0.118 to 0.467, and 87.5% of paired bootstrap samples favor the learned model. Thus, the direction is encouraging but not conclusive at a conventional 95% level. The uncertainty can be rationalized because the holdout contains only 22 molecules.

Figure 5 shows regression toward the training mean. The most active holdout molecule has observed pK_d 10.276 but predicted 6.359; its maximum Morgan/Tanimoto similarity to training is only 0.302. The largest errors occur at affinity extremes, consistent with limited chemical coverage. At the same time, high similarity is not sufficient for accuracy: one weak binder with similarity 0.955 is overpredicted by 2.21 pK_d units, plausibly reflecting assay variation or activity-cliff behavior.

Figure 5. Scaffold-holdout diagnostics expose where the selected model generalizes poorly

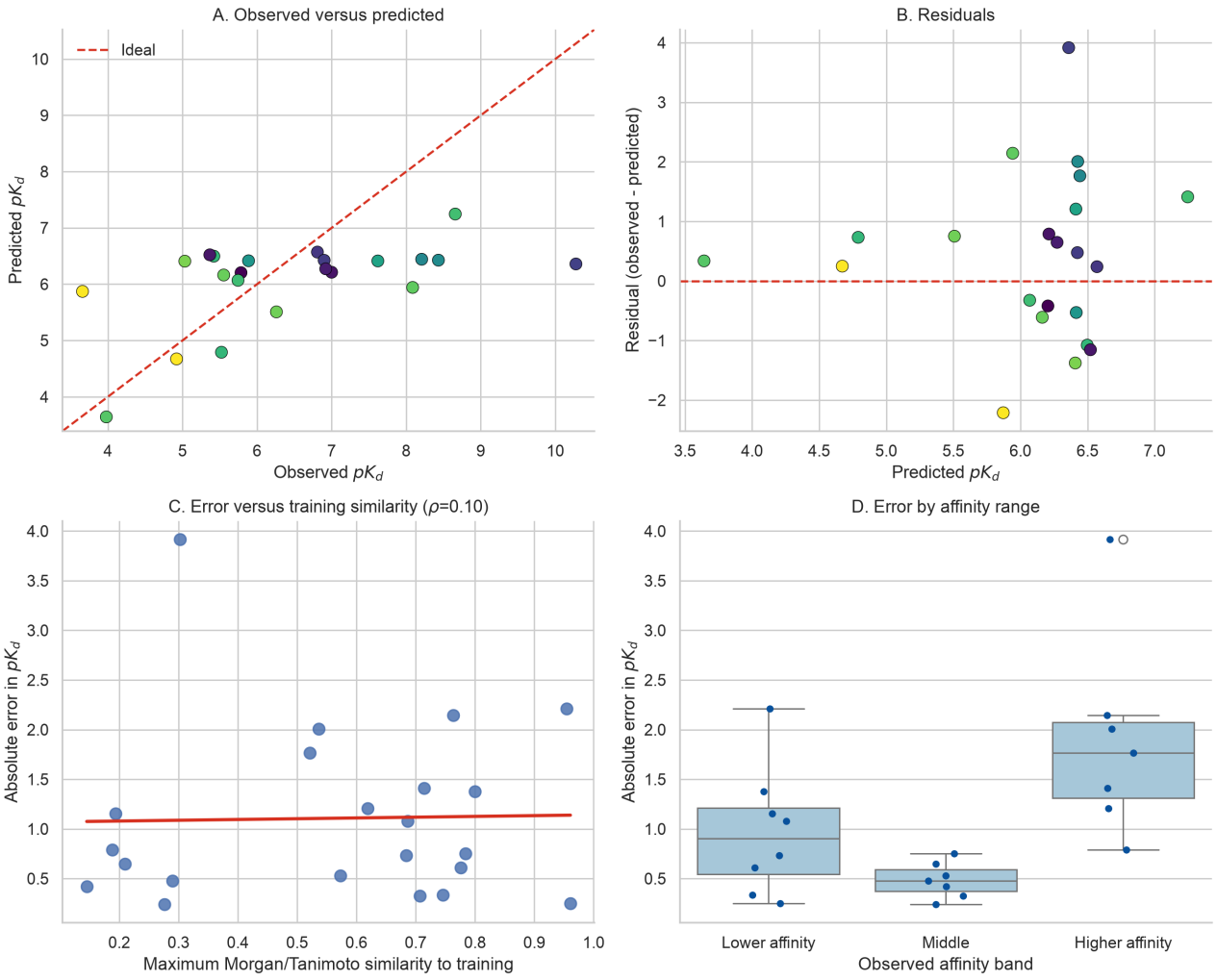


Figure 5. Final holdout diagnostics. The model ranks compounds moderately but compresses predictions toward the training center. Similarity helps characterize domain coverage but does not fully explain error.

5.3 Interpretability and sensitivity checks

Held-out permutation importance is diffuse. Morgan bits 1535 and 1325 have the largest mean importance, increasing MAE by 0.050 and 0.039 when shuffled, respectively; most selected bits have effects near zero relative to their permutation variability. The best ridge model is the combined representation. Its largest standardized coefficients include positive bit 1325 and negative bit 809; nitrile count has a negative coefficient. These are predictive associations in a hashed, correlated feature space, not an atom-level binding explanation.

The validation diagnostic is more decisive. Reusing the locked selected model on an ordinary 80/20 random split gives MAE 0.623, RMSE 0.855, $R^2 = 0.546$, and Spearman $\rho = 0.736$. Five test scaffold identities also occur in random-split training. The much stronger random result is evidence that analog leakage makes the task look easier. Finally, removing the two molecules with replicate range above 1.0 pK_d barely changes scaffold-holdout MAE (1.105 versus 1.108), so those two disagreements do not drive the headline result.

6 Discussion and Conclusions

Molecular structure contains predictive information for this frozen human GSK-3β dataset, but the useful conclusion is modest. A reduced Morgan fingerprint and random forest improve the scaffold-holdout point estimate over a median predictor and explain 21% of holdout variance. However, a 1.108-pK_d MAE is roughly a 12.8-fold affinity error, and the paired improvement interval crosses zero. This model is not accurate enough to replace experiments or rank structurally novel compounds with high confidence.

The data audit changes the scientific question as much as the algorithm does. The attractive same-platform design becomes infeasible when censored measurements are handled honestly. Falling back to the extended cohort preserves target and endpoint but mixes assays, constructs, and sources. The model may therefore learn laboratory variation together with chemistry. A larger homogeneous dataset, or an interval-regression model that uses the censored observations without treating them as points, would be a stronger next step.

The representation comparison is also informative. Interpretable descriptors come close to fingerprint performance, but neither representation is stable across scaffold folds. The 64-bit fold-local selector addresses the high-dimensionality concern without leaking validation labels, yet selected bits can change when the sample changes. Future work should report selector stability, acquire more exact measurements, and evaluate a time split or publication-blocked split in addition to scaffolds.

The central methodological lesson is that validation design dominates the headline. Random splitting cuts MAE almost in half, but that result answers an easier question about interpolation among familiar analogs. The scaffold result is less impressive and more relevant to genuinely new chemistry. Under that stricter definition, the pre-specified success criterion is met nominally, while the evidence remains uncertain and the practical error remains large.

A Compact Data Dictionary and Reproducibility Details

Table 4. Key fields in the derived modeling table. Audit and split fields are excluded from predictors.

Field	Role	Definition
canonical_smiles	Molecule key	Canonical isomeric SMILES after cleanup and counterion removal.
inchi_key	Molecule key	Standardized InChIKey used to audit identity.
pkd	Target	Median $9 - \log_{10}(K_d[\text{nM}])$ per standardized molecule.
replicate_n	Audit	Number of exact source measurements aggregated.
replicate_range_pkd	Audit	Maximum minus minimum pK_d among exact replicates.
scaffold_id	Split group	Chirality-aware Bemis-Murcko scaffold; never a predictor.
mol_wt, mol_logp, tpsa	Predictors	Calculated size, lipophilicity, and polar-surface descriptors.
hbd, hba, rings, fraction_csp3	Predictors	Hydrogen-bond, ring, and saturation summaries.
esol_logs	Predictor	Delaney model estimate of log solubility; not an experiment.
fr_*	Predictors	Thirteen RDKit functional-group counts.
morgan_0000-2047	Predictors	Chirality-aware radius-2 Morgan fingerprint bits.
article_dois, institutions	Audit	Provenance retained for traceability and excluded from models.

The analysis ran with Python 3.12.13, pandas 2.3.3, NumPy 2.5.2, scikit-learn 1.9.0, and RDKit 2026.03.5. The target TSV SHA-256 is

`fc388edea599b6490d37d8318b9216a88719cd1d156084bd039def7d87fa977b`

The executed notebook saves all raw-page hashes, the complete feature list, the deterministic split assignments, model parameters, metrics, predictions, and bootstrap summaries in `output/tables/analysis_manifest.json` and adjacent CSV files.

References

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